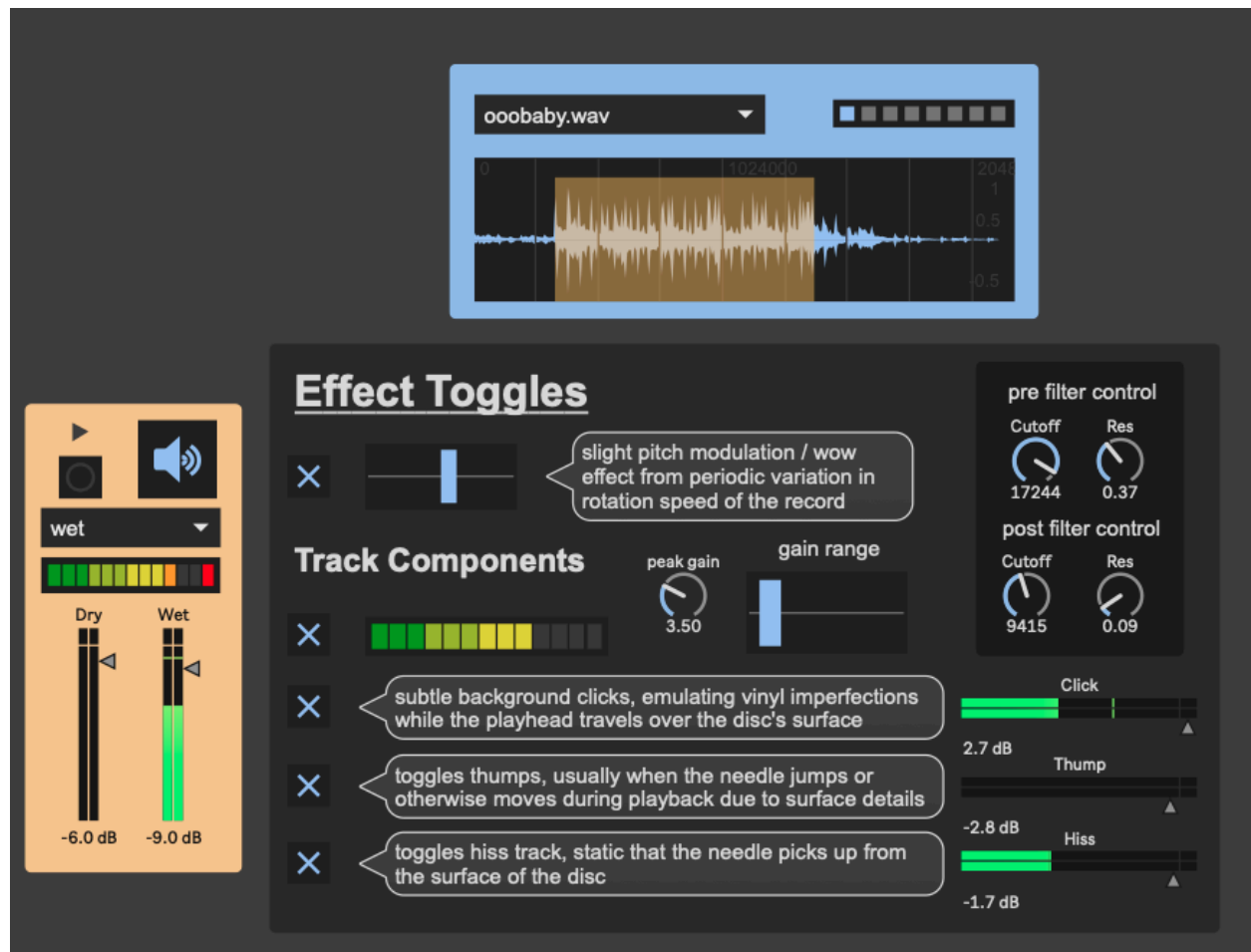


Computer Music Fundamentals Final Paper

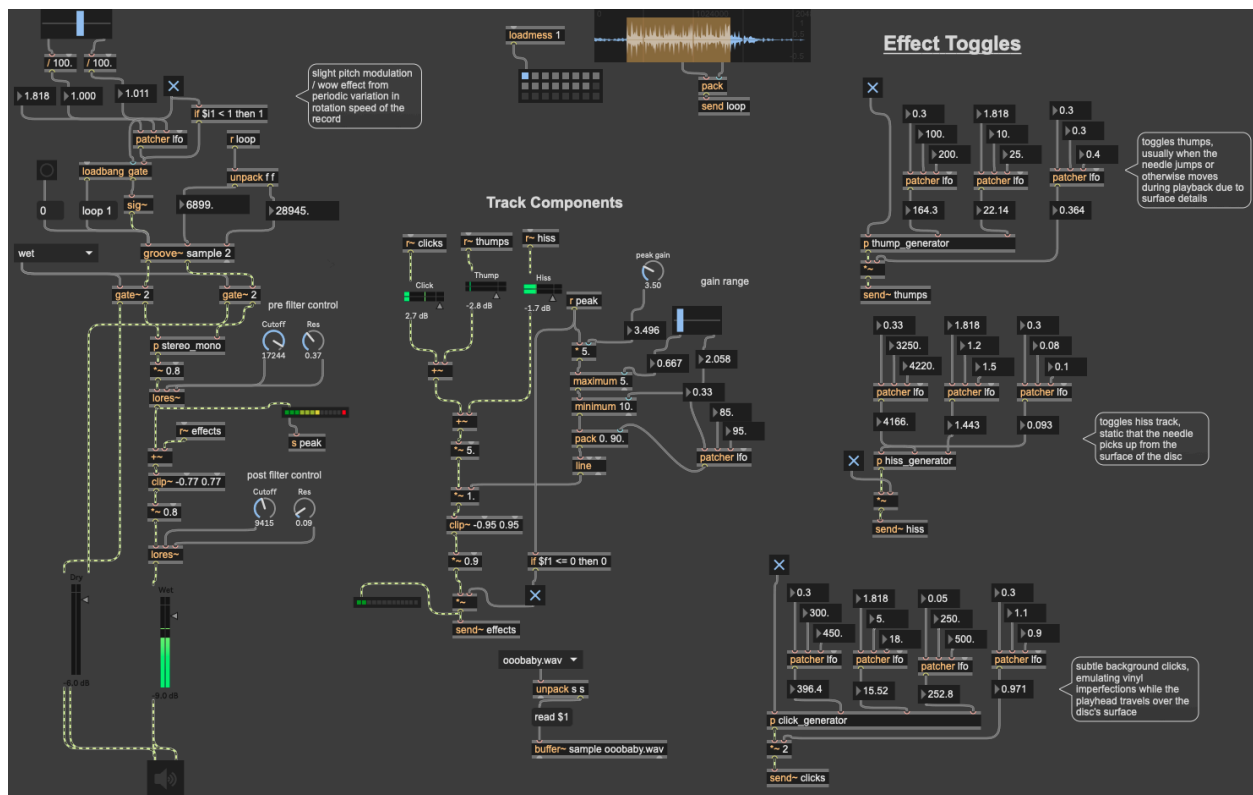
Seth Weintraub

My plan for the final project was to create a digital recreation of vinyl antiquing, to degrade samples or songs so they sound as if they have been played through a vinyl record. My patch simulates the physical and sonic imperfections of vinyl playback, to emulate the analog sound of the playback method. As the foundation for the work, I used a 2008 paper from the Journal of the Audio Engineering Society (AES) titled “Digital Audio Antiquing” by Välimäki et al., which outlines signal processing techniques to simulate historical recording media. Their case study on the vinyl LP formed the technical and conceptual backbone for my Max/MSP patch.



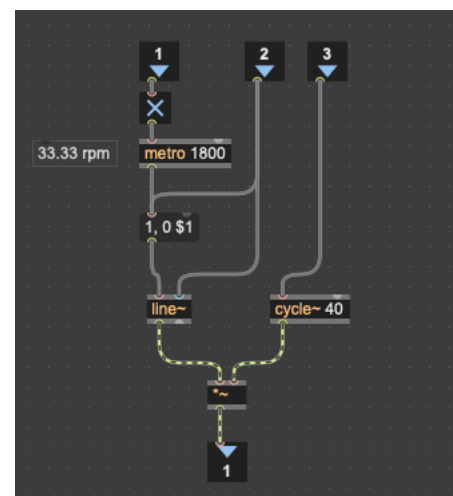
In the main patch view, the user interface is split into three sections. The first, with an orange background, provides playback controls and lets the user switch between the dry (no antiquing) and wet (antiqued) signal paths, monitor levels for each, along with controls to start and stop playback. The second section, in blue, is for sample selection. Here, the user can choose from a set of curation audio clips, including music and vocal samples, and define a loop range using the interactive waveform display. The third section is the central effects board, where users toggle and adjust vinyl degradation effects.

Among the effects are both real-time audio distortions and auxiliary signals mixed with the main audio to emulate vinyl playback. One of the more prominent effects, as it applies to the playback of the sample itself, is the pitch modulation or “wow” effect. This effect slightly alters the playback speed to simulate the subtle pitch variations caused by imperfections in vinyl rotation. A range slider allows the user to define how much the playback rate should be modulated, centered around the normal playback speed (first half is below 1, second half is above 1). Below that, toggles active three additional audio layers: background clicks (caused by dust and scratches on the vinyl surface), thumps (from physical disruptions, like the needle jumping), and static hiss. Each effect has a dedicated gain slider and metering. To further shape the signal, two low pass filters are present: one applied before the effects, and one after, to emulate the frequency constraints of analog vinyl playback.

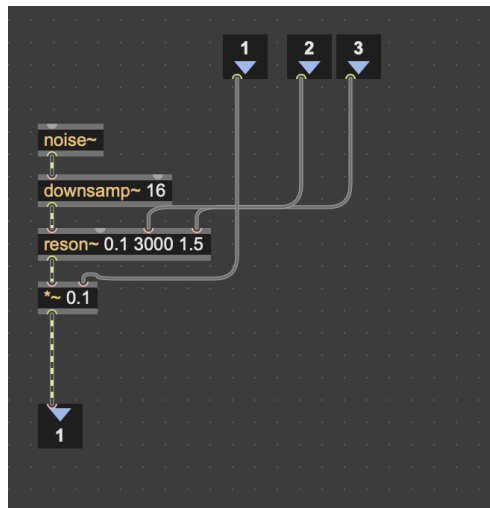


Inside the patch, each effect is handled in a separate subpatch. Most modulation is driven by LFOs and triggered events. The audio signal is first converted from stereo to mono by averaging the two channels, since early vinyl playback was monophonic. This mono singla is then processed through the first lowpass filter and passed along to the effects mixing section.

The thump effect is triggered using a metronome set to 1800 millisecond interval, reflecting a 33 ⅓ RPM record rotation. Each metronome tick triggers a decaying line function,

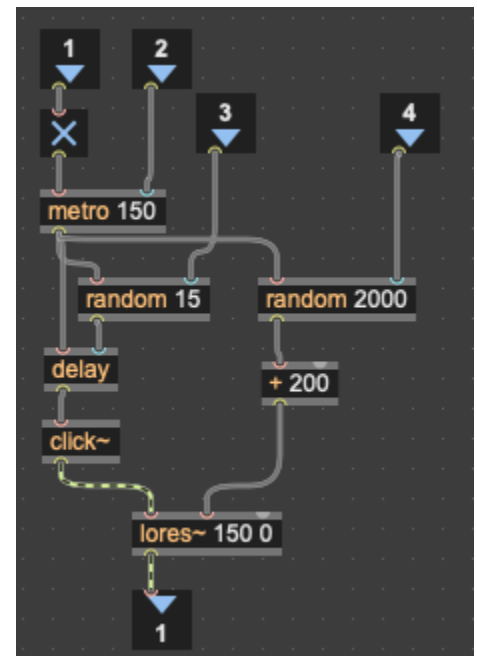


shaping a low frequency sine wave burst. This creates a low, resonant pulse similar to what is heard when the needle jumps.



For the hiss effect, downsampled white noise is passed through a bandpass filter centered around 3 kHz to isolate mid-high-frequency energy and avoid overwhelming the mix. The result is a subtle layer of analog-like static.

The click effect is produced using a metronome with randomized delay intervals. It triggers a [click~] object shaped by a short envelope and passed through a randomized lowpass filter to ensure that no two clicks sound exactly the same.



To introduce more dynamic interaction between the effects track and the sample being antiquated, I added a feature where the peak level of the clean signal modulates the intensity of the effects layer. A [meter~] object tracks the incoming audio's amplitude and outputs its peak value. This value is scaled, clamped to a customizable min-max range, and then used to control the gain of the summed effects before they are added back to the signal. This results in louder moments producing more intense vinyl degradation, creating natural-sounding crackles and saturation under transient peaks. This produces a slightly unstable, gritty sound, which helps to enhance the illusion of low-fidelity playback.

One of the main challenges was tuning effect parameters. Although the AES paper includes tables and specifications, converting these to my Max patch was not straightforward, leading to many values having to be adjusted by ear. Additionally, some DSP methods described in the paper, especially those using advanced math for LPC based filtered or physically modeled tracking errors, were too complex to directly implement in Max at this time. Instead, I focused on effects I could design intuitively, prioritizing listenability over exact technical replication.

Overall, the patch achieves my goal of degrading audio in a stylistically convincing way, giving clean digital samples the character of vintage vinyl playback. While simplified compared to the original models, the patch offers a flexible and sonically rich and musically expressive tool for digital antiquing, capable of adding fine degradations and character to a wide range of source material.

Sources

Välimäki, Vesa & Gonzalez, Sira & Kimmelma, Ossi & Parviainen, Jukka. (2008). Digital audio antiquing - Signal processing methods for imitating the sound quality of historical recordings. AES: Journal of the Audio Engineering Society. 56. 115-139.

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